

Spa-Bench training exposure and held-out spatial bindings

Physical State

Concept / argument	White cup	Striped cup	Brown cup	Spice container
Open	Observed	Withheld	Observed	Observed
Closed	Observed	Observed	Withheld	Observed
Upright	Observed	Observed	Observed	Withheld
Upside-down	Withheld	Observed	Observed	Observed

Relative Size

Concept / argument	Green block	Crushed Pepsi can	Red pencil sharpener	Juggling ball
Largest	Withheld	Observed	—	Observed
Smallest	Observed	—	Observed	Withheld

Referential Description

Concept / argument	Airplane toys	Crushed soda cans	Pencil sharpeners	Pens
Closest	Observed	Withheld	Observed	Observed
Furthest	Withheld	Observed	Observed	Observed
Between	Observed	Observed	Withheld	Observed

Counting

Concept / argument	Addition	Removal
0	—	Observed
1	Observed	Withheld
2	Observed	Observed
3	Withheld	Observed

- Observed binding
- Withheld binding
- Absent / N.A.

A withheld cell retains physical trajectory coverage under a direct or familiar instruction; only the spatial concept–argument binding is absent.

Ordinal Position — 2-object sequences

Concept / argument	Airplane toy with propellers	Blue tape ball	Green highlighter	Yellow pen
L1	Withheld	Observed	Withheld	Observed
L2	Observed	Withheld	Observed	Withheld

Ordinal Position — 3-object sequences

Concept / argument	Airplane toy with propellers	Blue tape ball	Green highlighter	Yellow pen
L1	Observed	Withheld	Observed	Withheld
L2	Absent	Absent	Absent	Absent
L3	Withheld	Observed	Withheld	Observed

Ordinal Position — 4-object sequences

Concept / argument	Airplane toy with propellers	Blue tape ball	Green highlighter	Yellow pen
L1	Withheld	Observed	Observed	Observed
L2	Observed	Withheld	Observed	Observed
L3	Observed	Observed	Withheld	Observed
L4	Observed	Observed	Observed	Withheld

Relational Placement

Concept / argument	Green pen + red block	Green pen + pink marker	Green pen + blue block	Red block + pink marker	Red block + blue block	Pink marker + blue block
Front	Withheld	Observed	Observed	Observed	Observed	Observed
Behind	Withheld	Observed	Observed	Observed	Observed	Observed
Left	Withheld	Observed	Observed	Observed	Observed	Observed
Right	Withheld	Observed	Observed	Observed	Observed	Observed